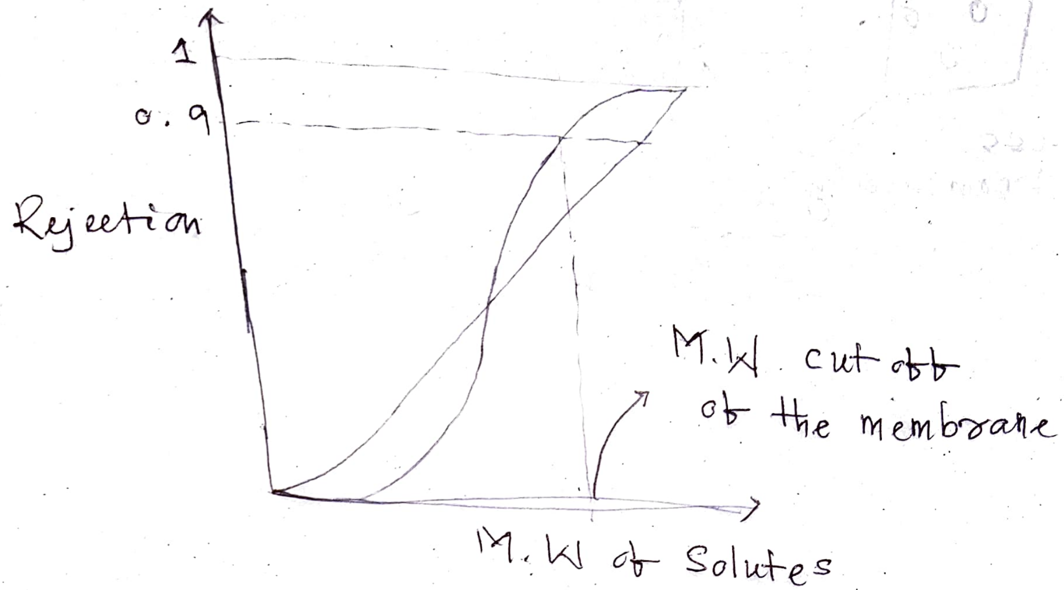
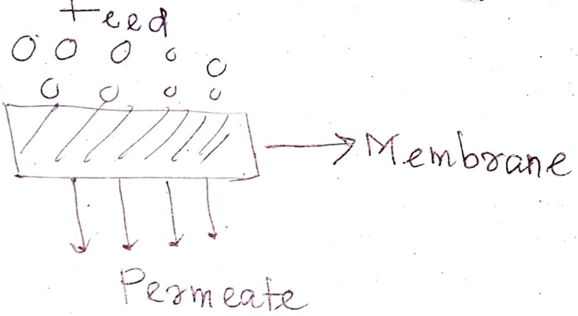
Mass ~~Heat~~ Transfer

Membrane Separation



$$\text{Rejection} = 1 - \frac{C_{permeate}}{C_{feed}}$$



Flux $\propto$ - pressure gradient

$u \propto -\nabla p \rightarrow$ Darcy's Law

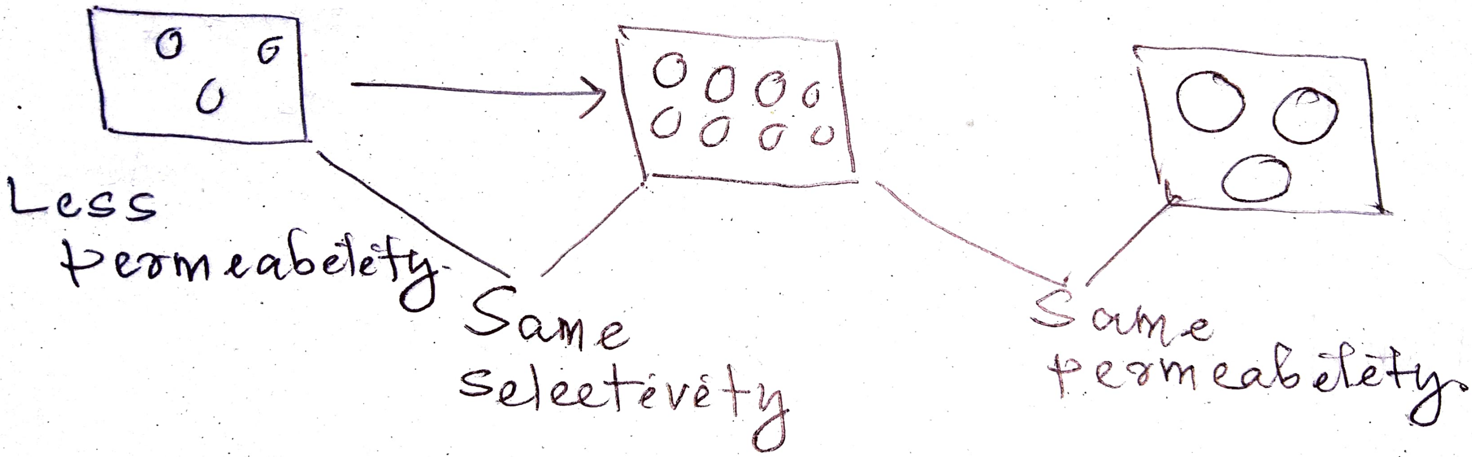
$\Rightarrow u = (\text{constant})(-\nabla p)$

$\Rightarrow u = \frac{-k}{\mu} \nabla p$

or $u = \underbrace{L_p}_{\text{Permeability of the membrane}} \Delta p$

or $u = \frac{\Delta p}{\mu k_m} \rightarrow \frac{1}{L_p}$

Porosity $\propto$ permeabelity



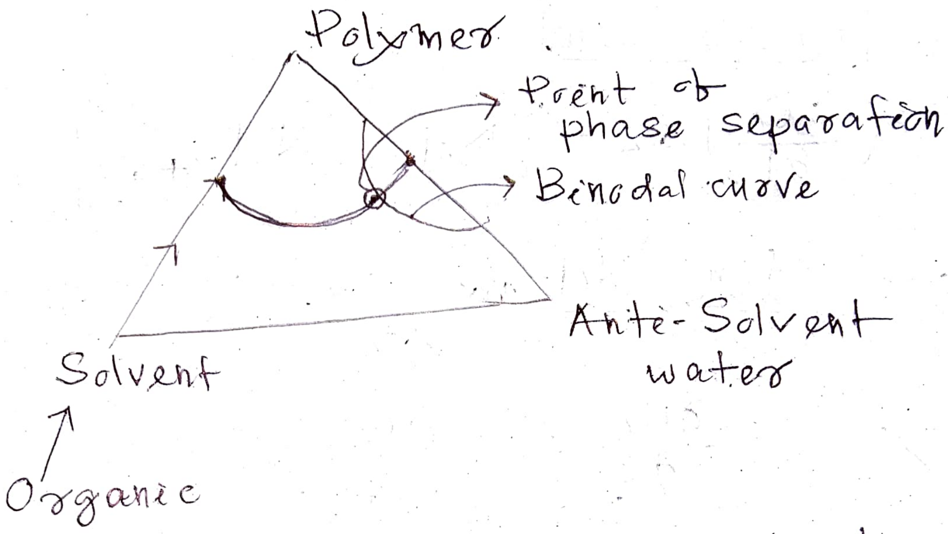
Membrane

Polymeric (Organic)
1. Not strong
2. Wide usages
3. Wide range of Selectivity

Ceramic (Inorganic)
Mechanically strong
limited use
not very selective

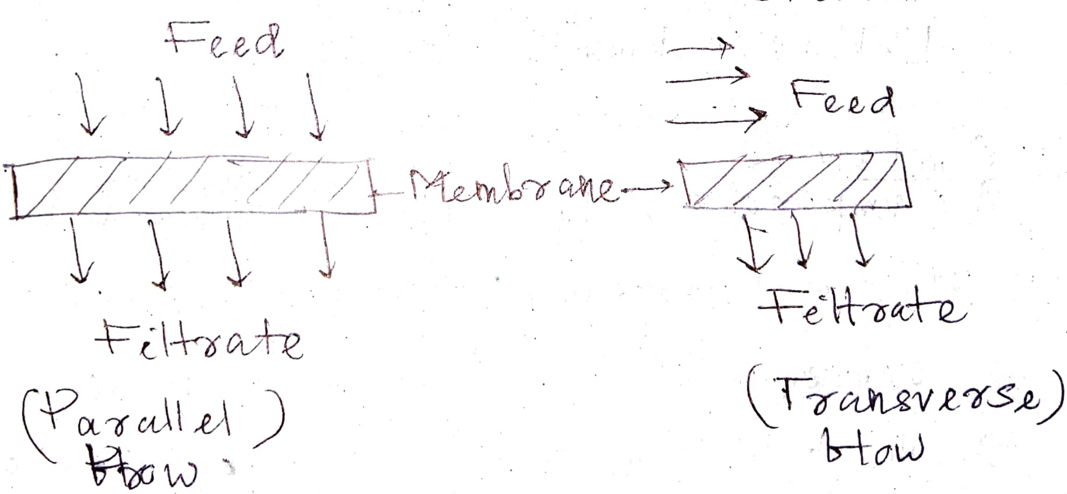
Phase Inversion:

Ternary Diagram



Left side of binodal curve → single phase not separable

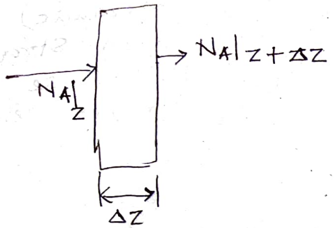
At & Right side → Phase separation starts



Batch Processes – Parallel & Transverse

Continuous Processes – Cross-flow

Diffusion:



Species concentration C_A

$\boxed{\text{Fick's Law}}$

$$\left(\frac{\partial C_A}{\partial t}\right)\Delta z = \left[N_A|_z - N_A|_{z+\Delta z}\right]$$

$$\Rightarrow \lim_{\Delta z \to 0} \frac{N_A|_{z+\Delta z} - N_A|_z}{\Delta z} = -\frac{\partial C_A}{\partial t}$$

$$\Rightarrow \frac{\partial N_A}{\partial z} = -\frac{\partial C_A}{\partial t}$$

$$\Rightarrow \frac{\partial C_A}{\partial t} = -\nabla . N_A$$

Fick's 1st Law:

Diffusive flux $\propto \nabla C_A$

Limitations

Assumptions: $\Rightarrow N_A = -N_B$ (equimolar counter diffusion)

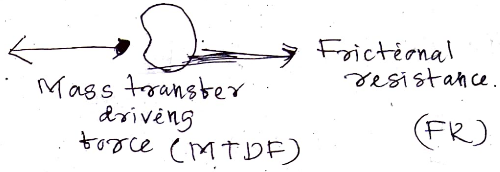
→ It is applicable ~~Binary~~ to only binary system

→ concentration of A is very dilute

→ the film should be stagnant / shouldn't be moving.

Stephene-Maxwell Law

Diffusion is a molecular phenomenon.



At molecular level

MTDF = FR

MT driving force

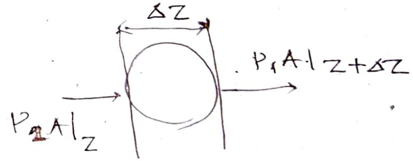
$$F_t = -\left(\frac{\partial \mu_i}{\partial z}\right)_{T,P} = -\nabla . \mu_i|_{T,P}$$

$\mu_i = RT \ln y_i P + \mu^*$

P = Total pressure.

- Change in total energy = Frictional loss

Friction force:



[Force per vol.)] $P_1 A|_z - P A|_{z+\Delta z} \propto \rho_1 \rho_2 (u_1 - u_2) A\Delta z$

Taking $\lim_{\Delta z \to 0}$

$$-\frac{dP_1}{dz} \propto \rho_1 \rho_2 (u_1 - u_2)$$

Consider ideal gas $P = \rho RT$

$$\boxed{F_t = -\left(\frac{\partial \mu_i}{\partial z}\right)_{T,P} = -\frac{RT}{P_i}\frac{\partial P_i}{\partial z}}$$

$F_1 \propto \rho x_2 (u_1 - u_2)$ $\quad \rho = \frac{P}{RT}$

$$\Rightarrow \boxed{F_1 = -\frac{RT}{D} x_2 (u_1 - u_2)}$$

$D \sim \frac{(RT)^2}{\rho}$ - ~~(diffusive constant)~~ diffusion co-efficient

From Maxwell Equation

$$-\frac{dP_1}{dz} = K A_1 P_1 \rho_2 (u_1 - u_2)$$

so, $D = \frac{(RT)^2}{K A_1 P}$

Multi-components

$$F_i = -RT \sum_{j=1}^{n-1} \frac{x_j}{D_{ij}} (u_i - u_j)$$

Equating the forces

$$-\frac{\partial \mu_i}{\partial z}\Big|_{T,P} = -RT \sum_{j=1}^{n-1} \frac{x_j}{D_{ij}} (u_i - u_j)$$

$$= -\frac{RT}{P}\frac{\partial p}{\partial z}$$

$$= -\frac{RT}{C_i}\frac{\partial C_i}{\partial z}$$

$C_i = x_i C_T$
$C_T = \sum C_i$
$N_i = C_i u_i$

On solving

$$\boxed{-\frac{\partial C_i}{\partial z} = \sum_{j=1}^{n-1} \frac{1}{C_T D_{ij}} (C_j N_i - C_i N_j)}$$

For the binary system

$$-\frac{\partial C_A}{\partial z} = \frac{1}{C_T D_{AB}} (C_B N_A - C_A N_B)$$

$$-\frac{\partial C_B}{\partial z} = \frac{1}{C_T D_{BA}} (C_A N_B - C_B N_A)$$

Stagnant component $N_B = 0$

$$-\frac{\partial C_A}{\partial z} = \frac{1}{C_T D_{AB}} C_B N_A$$

For dilute solⁿ

$C_A << C_B$

$C_A + C_B \simeq C_B$

$\Rightarrow C_T \simeq C_B$

$$\therefore -\frac{\partial C_A}{\partial z} = \frac{N_A}{D_{AB}}$$

$$\Rightarrow \boxed{N_A = -D_{AB}\frac{\partial C_A}{\partial z}} \quad \text{Fick's Law.}$$

~~Diffusion in Gases~~

Kinetic Theory of Gases

$$\text{Flux} = -\frac{1}{3}\tilde{v} l \nabla C + C v^{\circ}$$

Let the gas molecules be rigid spherical.

$-\frac{1}{3}\tilde{v} l \nabla C \longrightarrow$ molecular diffusion

$C v^{\circ} \longrightarrow$ convection

$\tilde{v}$ = molecular scale velocity

l = mean free path

v° = bulk velocity

Cunningham & Williams (1980)

Diffusion in gases and

Comparing with Fick's Law

$$D \simeq \frac{1}{3} v \tilde{l} \qquad \text{——— (1)}$$

From statistical molecules

$$\bar{v} = \sqrt{\frac{8}{\pi} \frac{k_B T N_A}{M_W}} \qquad \text{——— (2)}$$

$$l = \frac{k_B T}{P \sqrt{2} \pi d^2} \qquad \text{——— (3)}$$

d = diameter of molecule

Ideal gas law

$$PV = N_A RT$$

$$\Rightarrow \frac{N_A P}{RT} = \frac{\text{constant}}{\text{Vol.}}$$

From (1), (2) and (3)

$$D = \left[\frac{2}{3}\left(\frac{k_B}{\pi}\right)^{3/2} N_A^{1/2}\right] \frac{T^{3/2}}{P d^2 M_W^{1/2}}$$

Diffusion in liquids:

Stokes - Einstein Law (Equation)

Assumptions

1. Dilute Sol^n - (Isolated single sphere)
2. Moving slowly in a "continuum" solvent

$\therefore$ Force $= \eta v$

η = friction coefficient
v = velocity
$\eta = 6\pi\mu r \rightarrow$ Stokes law

Force $= -\nabla \mu_p$ ← chemical potential defined per molecule (not mole)

$$\mu_p = \mu_p^0 + k_B T \ln x$$

$$x = \frac{c}{c + c_s} \; ; c << c_s$$

$$\Rightarrow x = \frac{c}{c_s}$$

$$\therefore \mu_p \approx \mu_p^0 + k_B T \ln\left(\frac{c}{c_s}\right)$$

So,

$$\nabla \mu_p = \left(\frac{k_B T}{c}\right) \nabla c = (6\pi\mu r) v$$

$$\therefore \boxed{\text{Flux} \equiv c v = -\left(\frac{k_B T}{6\pi\mu r}\right) \nabla c}$$

$$D = \frac{k_B T}{6\pi\mu r}$$

Validation of Stokes - Einstein Law

1. $\frac{r_{solute}}{r_{solvent}} > 5$

$\sim 1-5$) (Instead of 6π use 4π)

2. Low viscosity of the solvent

3. No Interaction between molecules

For high viscosity, $D \sim \frac{1}{\mu^{2/3}}$

Extreme high viscosity $D \sim \frac{1}{\mu^0}$

- In case of concentrated Solution:

$$D \sim \frac{k_B T}{6\pi\mu r}\left(1 + 1.5\phi + O(\phi^2) \text{ \& higher}\right)$$

ϕ = volume fraction related to the concentration.

Diffusion in porous medium:

$$D_{eff} \sim [\varepsilon\, D_{bulk}]$$

$$D_{eff} < D_{bulk}$$

Comparision of diffusion coefficients:

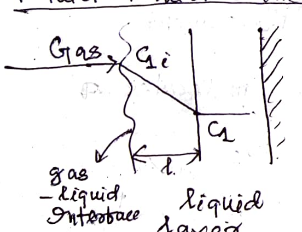
Phase	Magnitude (m^2/s)	Temp	Press.	dia	Visc
G	$10^{-6} - 10^{-5}$	$T^{3/2}$	$\frac{1}{P}$	$\frac{1}{d^2}$	μ (small)
L	$10^{-11} - 10^{-9}$	T	small	$1/d$	$1/\mu$ $1/\mu^{2/3}$ (for high viscosity)

Measuring Methods:

Dynamic Scattering

Speed Echo, Interferometer

Fluid-Fluid Interface:



$$N = k\,(C_{1i} - C_1)$$

non-dimensional form

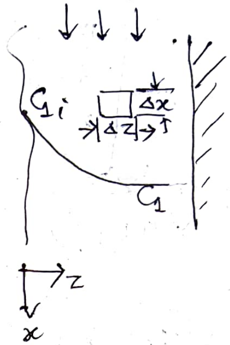
$$Sh = \frac{k}{D} \text{ (characteristic length scale)}$$

$$N = J = \frac{D}{l}(C_{1i} - C_1)$$

(Fickian diffusion)

$$\boxed{k = \frac{D}{l}}$$

The Penetration theory:



$$0 = (j\,\Delta x)\big|_z - (j\,\Delta x)\big|_{z+\Delta z} + (v\,c\,\Delta z)\big|_x - (v\,c\,\Delta z)\big|_{x+\Delta x}$$

$$\Rightarrow \lim_{\substack{\delta x \to 0 \\ \delta z \to 0}} \quad 0 = -\frac{\partial j}{\partial z} - \frac{\partial (vc)}{\partial x}$$

$$\Rightarrow \frac{\partial j}{\partial z} = -\frac{\partial (vc)}{\partial x}$$

$$j = -D\frac{\partial c}{\partial z}$$

$$\therefore \boxed{D\frac{\partial^2 c}{\partial z^2} = \frac{\partial (vc)}{\partial x}} \quad \text{——— (1)}$$

B.C

(a) $x = 0, \quad C = C_1$

(b) $z = 0, \quad C = C_{1i}$

(c) $z \to \infty, \quad c = C_1$

Solution of (1)

$$\frac{C_1}{(C_{1i} - C_1)} = 1 - \operatorname{erf}\left[\frac{z}{4Dx/v}\right]$$

So, flux @ interface

$$N\big|_{z=0} = \sqrt{\frac{Dv}{\pi x}}\,(C_{1i} - C_1)$$

$$v = v_{max}$$

$$k = \left(\frac{Dv}{\pi x}\right)^{1/2} = \text{mass transfer coefficient.}$$

Surface Renewal theory:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial z^2}$$

B.C
$t=0,\ C=C_1$
$z=0,\ C=C_{1i}$
$z\to\infty,\ C=C_1$

$$\frac{C_1}{C_{1i}-C_1} = 1-\operatorname{erf}\left(\frac{z}{\sqrt{4Dt}}\right)$$

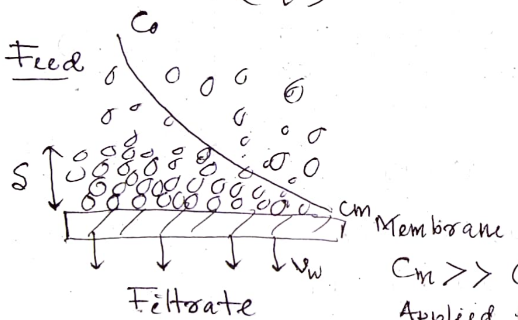
$$N\big|_{z=0} = \left(\frac{D}{\pi t}\right)^{1/2}(C_{1i}-C_1)$$

$$N_{t,avg} = \frac{\int_0^\infty E(t)\,N_t\,dt}{\int_0^\infty E(t)\,dt}$$

On solving

$$N_{avg} = \left(\frac{D}{\tau}\right)^{1/2}(C_{1i}-C_1)$$

$$k = \left(\frac{D}{\tau}\right)^{1/2}$$



$C_m >> C_0$

Applied Pressure > Osmotic Pressure

Concentration Polarization

$C_m\uparrow$ Osmotic Pressure $\pi\uparrow$ · $v_w\downarrow$

$(\Delta p - \Delta\pi)$

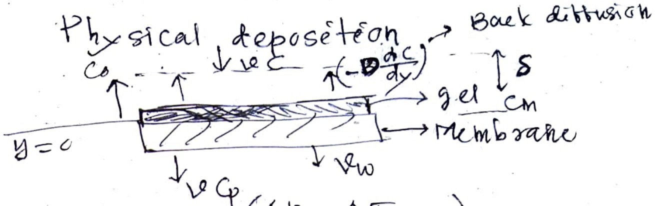
$V_w \propto$ driving force

$C_m >$ solubility limit

Precipitation

gel formation
(extra resistance)

Physical deposition



$$\therefore\ v_w = \left(\frac{\Delta p - \Delta\pi}{\mu(R_m+R_g)}\right)$$

R_m = Membrane hydraulic resistance

R_g = gel resistance

Viscosity is generally a function of concentration

$\mu \sim \mu(C)$

$C_m\uparrow\quad \mu\uparrow\quad v_w\downarrow$

C_p = filtrate Conc.

D = Back diffusivity

Net solute flux balance @ $y=0,\ C=C_0$
@ SS

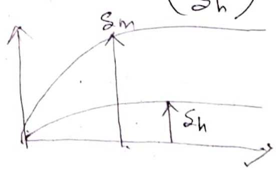
$$vC - \left(+D\frac{dC}{dy}\right) = vC_p$$

$$\Rightarrow D\frac{dC}{dy} = +v(C-C_p)$$

$$\Rightarrow \int_{C_m}^{C_0}\frac{dC}{(C-C_p)} = \frac{+v}{D}\int_0^{\delta_m} dy \qquad (v = \text{constant})$$

$$\Rightarrow \boxed{\ln\frac{C_0-C_p}{C_m-C_p} = \frac{+v}{D}\,\delta_m}$$

$$\delta_c \sim \left(\frac{\delta_m}{\delta_h}\right)^{1/2} \qquad \delta_m >> \delta_h$$



So, v is considered as constant

$v = -v_w$

$$\Rightarrow \frac{C_m-C_p}{C_0-C_p} = \exp\left(\frac{v_w}{D}\,\delta_m\right)$$

From Film theory $K = D/\delta$

$$\frac{C_m-C_p}{C_0-C_p} = \exp\left(\frac{v_w}{K}\right)$$

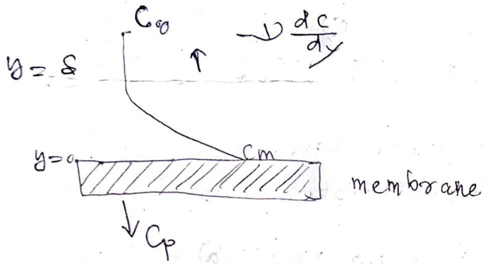
$$v_w = L_p(\Delta p - \Delta\pi)$$

$$\pi = \underbrace{(\text{constant})}_{b} \cdot C$$

$$\pi|_{\text{surface}(m)} = b \cdot C_m$$

$$\pi|_{\text{filtrate}(p)} = b\, C_p$$

$$\Delta\pi = b(C_m - C_p)$$



Flux balance

$$v_w C + D\frac{dc}{dy} = v_w C_p$$

On solving in $y \in [0, \delta]$

$$\boxed{\exp\left(\frac{v_w}{k}\right) = \left(\frac{C_m - C_p}{C_0 - C_p}\right)} \qquad (1)$$

$$k = D/\delta$$

Unknowns are v_w, C_m, C_p

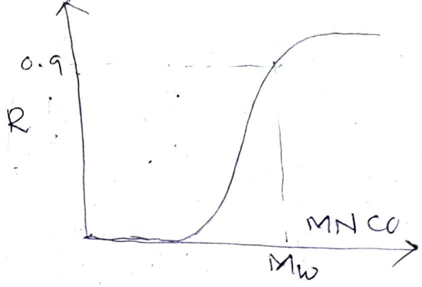
$$\boxed{v_w = L_p(\Delta p - \Delta\pi)} \qquad (2)$$

where $\pi = bC \Rightarrow \Delta\pi = b(C_m - C_p)$

$$b = \frac{RT}{M_w}$$

Real retention, $\boxed{R_r = 1 - \frac{C_p}{C_m}}$ (3)

(membrane characteristics)



Combining (1), (2) and (3)

$$\exp\left(\frac{L_p(\Delta p - \Delta\pi)}{k}\right) = \left(\frac{R_r C_m}{C_0 - C_p}\right)$$

$$\Rightarrow \exp\left(\frac{L_p}{k}(\Delta p - bR_r C_m)\right) = \left(\frac{R_r C_m}{C_0 - C_p}\right)$$

$$v_w = L_p(\Delta p - bR_r C_m)$$

$$\Rightarrow \Delta p - bR_r C_m = \frac{v_w}{L_p}$$

$$\Rightarrow C_m R_r = \frac{1}{b}\left(\Delta p - \frac{v_w}{L_p}\right)$$

$$\therefore \frac{\Delta p - \frac{v_w}{L_p}}{(C_0 - C_p)} = \exp\left[\frac{L_p}{k}\Delta p - b\left(\Delta p - \frac{v_w}{L_p}\right)\right]$$

Here $C_m = \frac{\Delta\pi}{bR_r}$; $C_p = \frac{(1-R_r)}{R_r}\frac{\Delta\pi}{b}$

Putting the values we get

$$\Rightarrow \boxed{v_w = L_p\left[\Delta p - bR_rC_0 \frac{\exp(v_w/k)}{R_r + (1-R_r)\exp(v_w/k)}\right]}$$

$$\Rightarrow v_w = L_p\left[\Delta p - \frac{bC_0}{\left(\frac{1-R_r}{R_r}\right) + \exp(-v_w/k)}\right]$$

Mass transfer coefficient (k)
(Sherwood number correlation)

$$\boxed{Sh = \frac{kL}{D}}$$

Parallel plate geometry

$$Sh = 1.86\left(Re\,Sc\,\frac{d_e}{L}\right)^{1/3} \quad \text{(Laminar)}$$

($Re \leq 2100$)

Cylindrical tube

$$Sh = 1.65\left(Re\,Sc\,\frac{d}{L}\right)^{1/3} \quad (\;"\;)$$

Turbulent

$$Sh = 0.023\,Re^{0.8}\,Sc^{0.3}$$

(1) Ideal/hypothetical

no concentration polarization

$C_m \sim C_0$

$$\frac{V_w}{k} \sim 0$$

$$\Rightarrow \exp\left(\frac{V_w}{k}\right) \sim 1$$

$$\therefore \quad V_w = L_p\left[\Delta p - \frac{b R_r C_0}{1}\right]$$

$$\Rightarrow V_w = L_p\left[\Delta p - b R_r C_0\right]$$

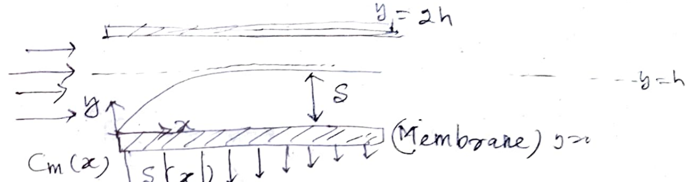
(2) Low polarization:

$$\boxed{\exp(V_w/k) = 1 + \frac{V_w}{k}} + O \text{ (higher order)}$$

$$\therefore \quad V_w = L_p\left[\Delta p - \frac{b R_r C_0\left(1+\frac{V_w}{k}\right)}{R_r + (1-R_r)\left(1+\frac{V_w}{k}\right)}\right]$$

(3) Low polarization and completely rejecting membrane ($R_r \sim 1$)

$$V_w = L_p\left[\Delta p - b\,C_0\left(1+\frac{V_w}{k}\right)\right]$$

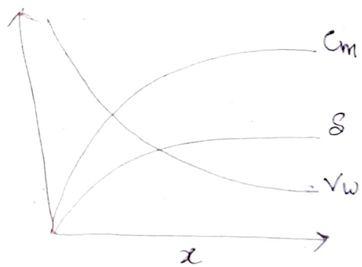


hydrodynamic BL,

$L_{ent} \sim 0.05\,Re$

→ Initially, C_p concentration is high as the membrane resistance is low.

As the resistance increases $C_p \downarrow$ gradually



$$u\frac{\partial c}{\partial x} + v\frac{\partial c}{\partial y} = D\left[\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2}\right]$$

$$\Rightarrow \vec{u}\cdot\nabla C = \nabla\cdot(D\nabla C)$$

Here

$$\frac{d^2 c}{dx^2} = 0 \quad \text{as} \quad \varepsilon = \frac{d_e}{L} \ll 1$$

$$v = -V_w(x)$$

$$D\frac{d^2 c}{dy^2} = u\frac{\partial c}{\partial x} + v\frac{\partial c}{\partial y}$$

For parallel plate:

$$u = \frac{3}{2} u_0 \left[\cancel{\frac{y}{h} - \left(\frac{y}{2h}\right)^2} \right] \quad u = \frac{3}{2} u_0 \left[1 - \left(\frac{y-h}{h}\right)^2 \right]$$

u_0 = cross-sectional velocity

Cylindrical

$$u = 2 u_0 \left[1 - \left(\frac{r}{R}\right)^2 \right]$$

$$u = \frac{3u_0}{2} \left[1 - \left(\frac{y-h}{h}\right)^2 \right]$$

$O(v) << O(u)$ $\quad [v_w << u_0]$

transmembrane pressure drop

axial flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{u_0}{L} \sim \frac{v_w}{h} \Rightarrow v_w \sim \frac{h}{L} u_0$$

$$\sim \varepsilon u_0$$

Here $v_w << u_0$ so the leaking effect doesn't affect the velocity profile

Region of interest $y \to 0$

If $y << 1$, $\quad u \simeq \frac{3u_0}{h} y$

$O(y^2) \approx$ small

$$\therefore u \frac{\partial c}{\partial x} + v \frac{\partial c}{\partial y} = D \frac{\partial^2 c}{\partial y^2} \quad \text{becomes}$$

$$\frac{3u_0}{h} y \frac{\partial c}{\partial x} - v_w \frac{\partial c}{\partial y} = D \frac{\partial^2 c}{\partial y^2}$$

BC

$x = 0$, $c = c_0$

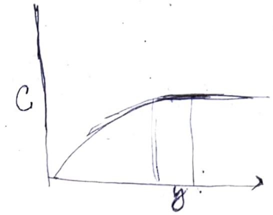
$y = h$, $c = c_0$

@ $y = 0$ (wall)

$$v_w c + D \frac{\partial c}{\partial y} = v_w c_p$$

Similarity technique

$$\eta = y/\delta$$



Evaluating at $y = \delta$

$$\frac{\partial c}{\partial y} = 0$$

Scaling Analysis

$$\frac{3u_0}{h} \delta \frac{\Delta c}{x} \sim D \frac{\Delta c}{\delta^2}$$

$$\Rightarrow \delta \sim \left(\frac{D x h}{3u_0}\right)^{1/3}$$

$$\eta = \frac{y}{\delta} = y \left(\frac{D x h}{3u_0}\right)^{-1/3}$$

$$\frac{\partial c}{\partial x}, \frac{\partial c}{\partial y}, \frac{\partial^2 c}{\partial y^2} \to \frac{\partial c}{\partial \eta}; \ \frac{\partial^2 c}{\partial \eta^2}$$

$$\frac{\partial c}{\partial y} = \frac{\partial c}{\partial \eta} \frac{\partial \eta}{\partial y}$$

$$\frac{\partial c}{\partial x} = \frac{\partial c}{\partial \eta} \frac{\partial \eta}{\partial x}$$

$$\frac{\partial^2 c}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial c}{\partial y}\right)$$

$$\frac{\partial^2 c}{\partial y^2} = \frac{\partial^2 c}{\partial \eta^2} \left(\frac{\partial \eta}{\partial y}\right)^2 + \frac{\partial c}{\partial \eta} \frac{\partial^2 \eta}{\partial y^2}$$

$$\frac{3u_0}{h} y \left(\frac{\partial c}{\partial \eta} \frac{\partial \eta}{\partial x}\right) - v_w \left(\frac{\partial c}{\partial \eta} \frac{\partial \eta}{\partial y}\right)$$

$$= D \left[\frac{\partial^2 c}{\partial \eta^2} \left(\frac{\partial \eta}{\partial y}\right)^2 + \frac{\partial c}{\partial \eta} \frac{\partial^2 \eta}{\partial y^2} \right]$$

$$\Rightarrow \frac{3u_0}{h} y \left(-\frac{y}{3} \left(\frac{x^4 h D}{3u_0}\right)^{-1/3} \frac{\partial c}{\partial \eta} \right) - v_w \left(\frac{D x h}{3u_0}\right)^{-1/3} \frac{\partial c}{\partial \eta}$$

$$= D \left(\frac{x D h}{3u_0}\right)^{-2/3} \frac{\partial^2 c}{\partial \eta^2} + 0$$

$$\left[-\frac{u_0 y^2}{h}\left(\frac{x h D}{3u_0}\right)^{-1/3} - v_w\left(\frac{D x h}{3u_0}\right)^{-1/3}\right]\frac{\partial c}{\partial \eta} = \left(\frac{D x h}{3 u_0}\right)^{-2/3} D\frac{\partial^2 c}{\partial \eta^2}$$

$$\Rightarrow \left[-\frac{u_0 y^2}{h x}\left(\frac{x D h}{3u_0}\right)^{1/3} - v_w\left(\frac{D x h}{3u_0}\right)^{1/3}\right] = D\frac{\partial^2 c}{\partial \eta^2}$$

$$\Rightarrow -\frac{y^2}{1}\left(\frac{D x h}{3u_0}\right)^{-2/3} - v_w\left(\frac{h x}{u_0 D^2}\right)^{1/3} = \frac{\partial^2 c}{\partial \eta^2}$$

$$\Rightarrow \left[-\eta^2 - v_w\left(\frac{h x}{u_0 D^2}\right)^{1/3}\right]\frac{\partial c}{\partial \eta} = \frac{\partial^2 c}{\partial \eta^2}$$

$$\delta = \left(\frac{h D x}{3u_0}\right)^{1/3}$$

$$\boxed{v_w \sim \frac{1}{\delta}} \sim \frac{1}{x^{1/3}}$$

$v_w x^{1/3} \longrightarrow$ constant

$v_w \left(\frac{h x}{u_0 D^2}\right)^{1/3} \longrightarrow A =$ constant

$$[-\eta^2 - A]\frac{\partial c}{\partial \eta} = \frac{\partial^2 c}{\partial \eta^2}$$

Integrating

$$\frac{\partial c}{\partial \eta} = K_1 \exp\left(-\frac{\eta^3}{3} - A\eta\right)$$

Again

$$C(\eta) = K_1\int_0^\eta \exp\left(-\frac{\eta^3}{3} - A\eta\right)d\eta + K_2$$

B.C. $y = h$. $(y \to \infty)$

$\delta/h << 1$

$\Rightarrow \eta \to \infty \Rightarrow C/C_0 = 1$

$\eta \to 0$

$$\frac{d(C/C_0)}{d\eta} + A\left(\frac{C_m}{C_0}\right)R_r = 0$$

$$v_w C|_{y=0} + D\frac{\partial c}{\partial y} = v_w C_p$$

$C|_{y=0} = C_m$

$$R_r = 1 - C_p/C_m$$

So, $K_1 = \dfrac{-A R_r C_0}{1 - A R_r I}$

$$\left[K_2 = C_m = \frac{C_0}{1 - A R_r I}\right]$$

where $I = \int_0^\infty \exp\left(-\frac{\eta^3}{3} - A\eta\right)d\eta$

$$\rightarrow \frac{3u_0 y}{h}\frac{\partial c}{\partial x} - v_w\frac{\partial c}{\partial x} = D\frac{\partial^2 c}{\partial x^2}$$

@ $x = 0$, $C = C_0$

@ $y = 0$ (membrane surface), $v_w C|_{y=0} + D\frac{\partial c}{\partial y}\Big|_{y=0} = v_w C_p$

@ $y = \frac{\delta}{h} \to \infty$, $C = C_0$

Similarity variable

$$\eta = \frac{y}{\delta} = \left(\frac{D x h}{3u_0}\right)^{-1/3} y$$

$$(-\eta^2 - A\eta)\frac{\partial c}{\partial \eta} = \frac{\partial^2 c}{\partial \eta^2}$$

$$\Rightarrow \frac{\partial c}{\partial \eta} = K_1 \exp\left(-\frac{\eta^3}{3} - A\eta\right)$$

$$C = K_1\int_0^\eta \exp\left(-\frac{\eta^3}{3} - A\eta\right)d\eta + K_2$$

@ $\eta = 0$ $(y = 0)$

$$\frac{\partial c}{\partial \eta} + A\, C|_{y=0}\, R_r = 0$$

$C|_{y=0} = C_m$

$$V_w = L_p(\Delta p - \Delta \pi) \quad \pi \sim f(C|_{y=0})$$

$$\Delta\pi = \pi_m - \pi_p$$

$$= f(C|_{y=0}) - f(C_p)$$

$$= f(C|_{y=0}, C_p)$$

Solution Steps:

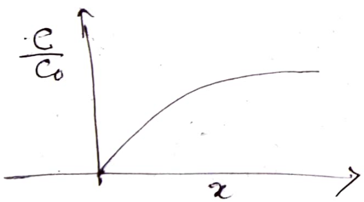
1. Guess the value of C_m ($C|_{y=0}$) $\sim f(x)$ or @ specific x

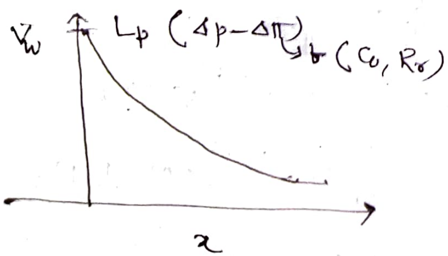
2. Calculate $V_w(x)$, A

3. Evaluate I and then $C(\eta)$ and $C(\eta=0)$ $\to$ $C|_{y=0}$ (C_m)

4. Compare the value of C_m from step 3 with step-1.

5. If difference is small, then choose a different x.





Fixed Suction/ Permeation:

$$Pe_w(\tilde{x}) = A\left(4 Re\, Sc \frac{d_e}{L}\right)^{1/3} \left(\frac{x}{L}\right)^{-1/3}$$

$Pe_w \to \frac{V_w d_e}{L}$ (non-dimensional wall Peclet no.)

$\tilde{x} = x/L$

$$Pe_w = V_w \left(\frac{h x}{u_o D^2}\right)^{1/3} \left(4 Re\, Sc \frac{d_e}{L}\right)^{1/3} \left(\frac{x}{L}\right)^{-1/3}$$

$$= V_w\left(\frac{h x}{u_o D^2}\right)^{1/3}\left(4 \frac{\rho u_o d_e}{\mu} \cdot \frac{\mu}{\rho D} \frac{d_e}{L}\right)^{1/3}\left(\frac{x}{L}\right)^{-1/3}$$

$$= \frac{V_w}{D}(4h\, d_e^2)^{1/3}$$

$$= \frac{V_w}{D}(d_e^3)^{1/3} \quad (\because d_e = 4h).$$

$$= \frac{V_w d_e}{D}$$

$$A = V_w\left(\frac{h x}{u_o D^2}\right)^{1/3}$$

$$=$$

$$\therefore \overline{Pe_w} = \int_0^1 Pe_w(\tilde{x})\, d\tilde{x}$$

$$= A\left(4 Re\, Sc \frac{d_e}{L}\right)^{1/3} \left[\frac{\tilde{x}^{2/3}}{2/3}\right]_0^1$$

$$\boxed{\overline{Pe_w} = 2.38\, A\left(Re\, Sc \frac{d_e}{L}\right)^{1/3}}$$

$$A = \frac{0.42\,\overline{P_{ew}}}{\left(Re\,Sc\,4e/L\right)^{1/3}}$$

→ Fixed suction means we are defining $\overline{v_w}$ (rate of permeation)

Mass transfer Coefficient

$$K\left(C|_{y=0} - C_0\right) = -D\,\frac{dC}{dy}\Big|_{y=0} \rightarrow K_1$$

K_2

$$\frac{dC}{d\eta}\Big|_{y=0}$$

$$\Rightarrow K\left(K_2 - C_0\right) = -D K_1 \left(\frac{u_0}{h x D}\right)^{1/3}$$

$$\frac{dC}{d\eta} = \frac{dC}{dy}\,\frac{dy}{d\eta}$$

or
$$\frac{dC}{dy} = \frac{dC}{d\eta}\,\frac{d\eta}{dy}$$

$$\frac{\partial C}{\partial y}\Big|_{y=0} = \frac{1}{\delta}\,\frac{\partial C}{\partial \eta}\Big|_{y=0} = \left(\frac{D x h}{3 u_0}\right)^{-1/3} \frac{\partial C}{\partial \eta}\Big|_{y=0}$$

$$K = -\left(\frac{K_1}{K_2 - C_0}\right)\left(\frac{u_0 D^2}{h x}\right)^{1/3} = \frac{1}{I}\left(\frac{u_0 D^2}{h x}\right)^{1/3}$$

$$Sh = \frac{k\,de}{D} = \frac{1}{I}\left(4 Re\,Sc\,\frac{de}{L}\right)^{1/3} x^{-1/3}$$

$$\therefore \overline{Sh} = \frac{1}{L}\int_0^L Sh(x)\,dx = \frac{2.38}{I}\left(Re\,Sc\,\frac{de}{L}\right)^{1/3}$$

In case of no-permeation ($A=0$)

$\overline{P_{ew}} = 0 \rightarrow (A=0)$

$$I = \int_0^{\infty} e^{-(\eta^3/3)}\,d\eta$$

$$\Rightarrow I = 1.288$$

$$\rightarrow \overline{Sh} = 1.85\left(Re\,Sc\,\frac{de}{L}\right)^{1/3}$$

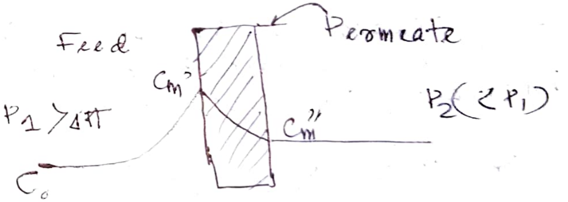
(rectangular impervious condition)

$$\rightarrow \overline{Sh} = 1.62\left(Re\,Sc\,\frac{de}{L}\right)^{1/3}$$

(Cylindrical tube channel)

Reverse Osmosis:

Solution Diffusion Theory



Darcy's Law: $v_w = L_p(\Delta p - \Delta\pi)$

$f(C_m', C_m'')$

Real retention: $R_r = 1 - \dfrac{C_m''}{C_m'} = f(L_p, \Delta P, \Delta\pi, B)$

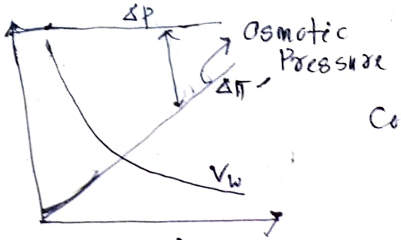
Solute flux balance: $v_w C_m'' = B\left(C_m' - C_m''\right)$

solute permeability

Unknowns: C_m', C_m'', v_w

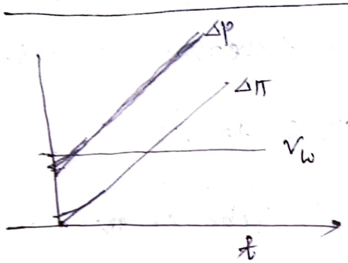
$\Delta p = P_1 - P_2$

①



Constant Pressure filtration Δp

② Constant rate filtration (V_w)



Δπ & Δp may not be linear.

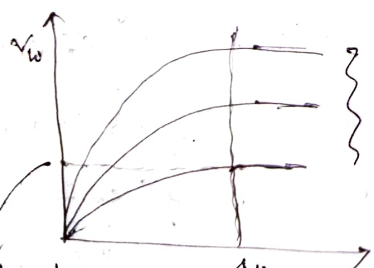
→ RO is mostly for small molecules.

Remarks

Case-1 is preferable because in the 2nd case it is very difficult to compress the increasing pressure.

Ultra and Micro filtration:

For larger molecules
(Proteins, enzymes etc.)



(limiting flux due to gel formation)

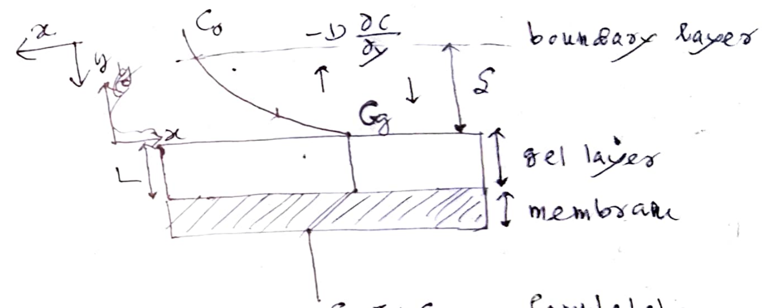
$$V_w = \frac{\Delta p}{\mu (R_m + R_g)}$$

R_g = Gel resistance
$= f(L)$.
L = Gel thickness
$= f(\Delta p)$

In case of limiting flux

$\boxed{R_g \gg R_m}$

($\frac{x}{\alpha + x}$ is constant if $x \gg \alpha$)



~~$k(C_0 - C_g) =$~~

$V_w C - D\frac{\partial c}{\partial y}$ ~~$V_w C_g$~~ $= (1-\varepsilon_g)\rho_g \frac{dL}{dt}$

→ $V_w C - D\frac{\partial c}{\partial y} = (1-\varepsilon_g)\rho_g \frac{dL}{dt}$

ε_g = gel porosity
ρ_g = gel density
L = gel thickness = L(t)

@ y = (edge of BL) $C = C_0$
$= 0$

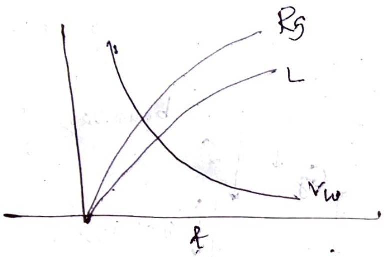
@ y = gel surface $C = C_g$
$= \delta$

$$\boxed{(1-\varepsilon_g)\,\rho_g \frac{dL}{dt} = V_w \frac{C_g - C_0 \exp(V_w/k)}{1-\exp(V_w/k)}}$$

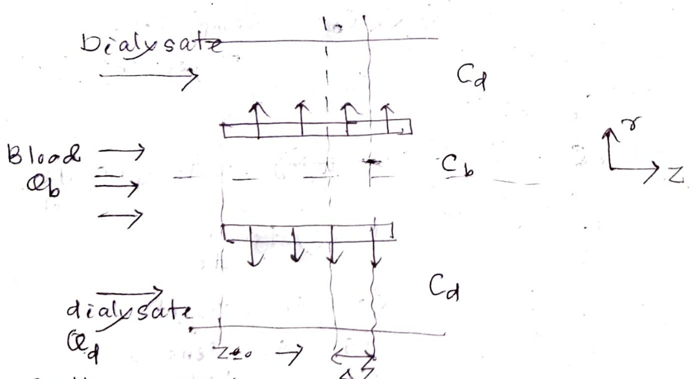
$$V_w = \frac{\Delta P}{\mu(R_m + R_g)}$$

where $R_g = \alpha(1-\varepsilon_g)\rho_g L$

α = Linear Resistance



Heamodialysis :



On the blood side

$$Q_b C_b|_z - Q_b C_b|_{z+\Delta z} = K(2\pi R\Delta z)(C_b - C_d) \quad (1)$$

On the dialyzer side

$$Q_d C_d|_z - Q_d C_d|_{z+\Delta z} = -K(2\pi R\Delta z)(C_b - C_d) \quad (2)$$

Since Q_b and Q_d are much high as compared to the flow across the membrane

$\Rightarrow$ Q_b and Q_d are assumed to be constant.

From (1) and (2)

$$Q_b \lim_{\Delta z\to 0} \frac{C_b|_z - C_b|_{z+\Delta z}}{\Delta z} + Q_d \lim_{\Delta z\to 0} \frac{C_d|_z - C_d|_{z+\Delta z}}{\Delta z} = 0$$

$$\Rightarrow Q_b \frac{dC_b}{dz} + Q_d \frac{dC_d}{dz} = 0$$

Integrating

$$Q_b C_b + Q_d C_d = \text{constant} = A \text{ (let)} \quad (3)$$

In general

$$Q_b C_{bi} + Q_d C_{di} = \text{constant.}$$

$i = 0, 1, 2 \ldots$

From (3) $\quad C_d = A - \left(\frac{Q_b}{Q_d}\right) C_b$

using in (1)

$$Q_b \frac{dC_b}{dz} = -(2\pi R K)\left(C_b + \frac{Q_b}{Q_d} C_b\right)$$

$$\Rightarrow \frac{dC_b}{C_b} = -\frac{2\pi R K}{Q_b Q_d}(Q_d + Q_b)\, dz$$

Integratins

$$\int_{C_{b0}}^{C_b} \frac{dC_b}{C_b} = -\frac{2\pi R K}{Q_b Q_d}(Q_d + Q_b)\int_0^z dz$$

$$\Rightarrow \ln \frac{C_b}{C_{b0}}$$

$$Q_b C_b + Q_d C_d = Q_b C_{bi} + Q_d C_{di}$$

$$\Rightarrow \quad C_d = C_{di} + \frac{Q_b}{Q_d}\left(C_{bi} - C_b\right)$$

We have

$$Q_b \frac{dC_b}{dz} = -k\,(2\pi R)\,(C_b - C_d)$$

$$\Rightarrow Q_b \frac{dC_b}{dz} = -k\,(2\pi R)\left(C_b - C_{di} - \frac{Q_b}{Q_d}(C_{bi} - C_b)\right)$$

$$\Rightarrow \int_{C_{bi}}^{C_{bo}} \frac{dC_b}{\left(1+\frac{Q_b}{Q_d}\right)C_b - C_{di} - \frac{Q_b}{Q_d}C_{bi}} = \frac{-2\pi R K}{Q_b}\int_0^l dz$$

$$\Rightarrow \ln\left[\frac{\left(1+\frac{Q_b}{Q_d}\right)C_{bo} - C_{di} - \frac{Q_b}{Q_d}C_{bi}}{\left(1+\frac{Q_b}{Q_d}\right)C_{bi} - C_{di} - \frac{Q_b}{Q_d}C_{bi}}\right]\frac{1}{\left(1+\frac{Q_b}{Q_d}\right)}$$

$$= \frac{-2\pi R K\, l}{Q_b}$$

$$\Rightarrow \ln\left[\frac{(1+Q_b/Q_d)\,C_{bo} - \left\{C_{di} + \frac{Q_b}{Q_d}C_{bi}\right\}}{(1+Q_b/Q_d)\,C_{bi} - \left\{C_{di} + \frac{Q_b}{Q_d}C_{bi}\right\}}\right] = -\left(1+\frac{Q_b}{Q_d}\right)\frac{k(2\pi R l)}{Q_b}$$

$$A_m = 2\pi R l$$

$$\frac{k\,(2\pi r l)}{Q_b} = \text{transfer units}$$

→ Clearance Cl

$$\underbrace{Q_b\,(C_{bi} - C_{bo})}_{\text{total amount of solute removed}} = \dot{M} = CL\,(C_{bi} - 0)$$

→ Dialysance: solute removal capacity

$$DL = \frac{\dot{M}}{C_{bi} - C_{di}}$$

If $C_{di} = 0$

$$CL = DL$$

Extraction Ratio

$$ER = \frac{DL}{Q_b} = \frac{\text{dialysance}}{\text{flow rate of impure blood}}$$

$$= \frac{C_{bi} - C_{bo}}{C_{bi} - C_{di}}$$